

MDE Attack Disruption Simulation

Attack path: Kali Linux (local, VMware) → Azure VM (MDE-onboarded)

1. Objective & Scope

Attack disruption only fires when Defender correlates **multiple high-confidence signals** into a single incident — it does not trigger on one isolated alert. With a single Azure VM + Kali, and no Defender for Identity / AD DS in the environment, the realistic outcomes are:

Outcome	Status	Notes
Device containment (endpoint isolated from the network by MDE)	✅ Achievable	Core focus of this simulation
User containment (account disabled/blocked tenant-wide)	❌ Out of scope	Requires Defender for Identity on a domain controller

Goal: build a multi-stage attack story from Kali, get MDE to correlate it into a high-confidence incident, and observe how far attack disruption goes without identity-layer signals in play.

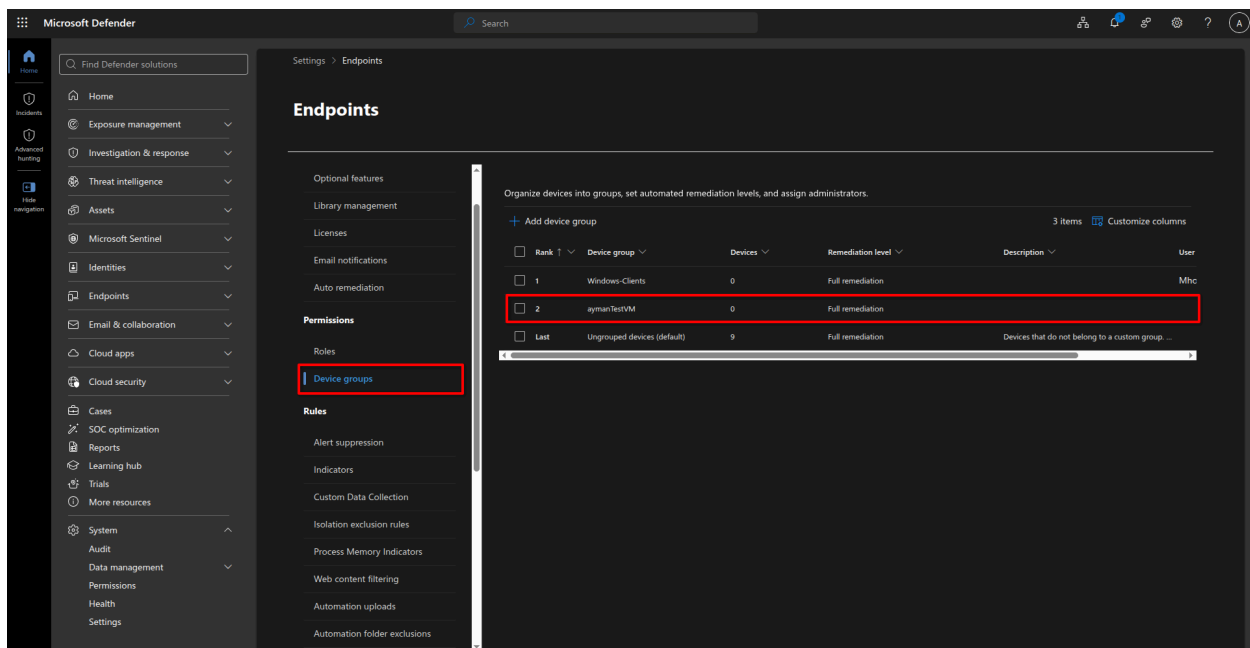
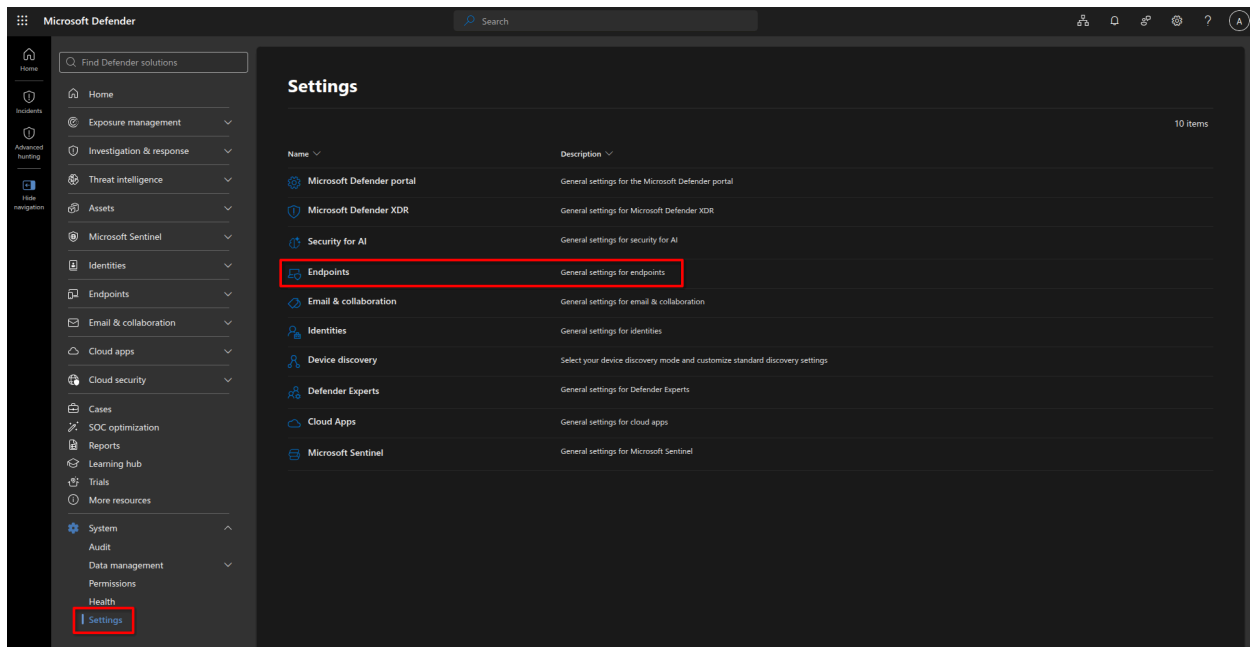
2. Prerequisites Checklist

Verify all of the following in the Defender portal **before** touching Kali.

2.1 Device group remediation level

Defender portal → System → Settings → Endpoints → Device groups

Confirm the device group containing the Azure VM is set to **"Full – remediate threats automatically."** Anything less (e.g. "No automated response") will block auto-containment even if detection fires correctly.



2.2 Sense agent version

On the Azure VM:

```
Get-Item "$env:ProgramFiles\Windows Defender Advanced Threat Protection\MsSense.exe" | Select-Object -ExpandProperty VersionInfo
```

Confirm the build is $\geq$ **10.8470** — the minimum version supporting Contain Device/User actions.

2.3 Defender AV operating state

```
Get-MpComputerStatus | Select AMRunningMode
```

Confirm it's at least in **Passive** or **EDR Block Mode**. Attack disruption itself works independent of AV state, but having the AV engine actively observing file/process activity improves signal richness.

2.4 Baseline check

[Defender portal](#) → [Assets](#) → [Devices](#) — confirm the Azure VM shows as **Active** and reporting normally before starting.

3. Phase 1 — Reconnaissance

From Kali, against the Azure VM's IP:

```
# Service/OS discovery
nmap -sV -O -Pn <azure_vm_ip>
```

This establishes the discovery stage of the attack story in Defender. Expect low-severity alerts such as "Suspicious network scanning activity" or similar reconnaissance detections — these don't trigger disruption on their own, but they matter for incident correlation in later phases.

Verify: [Defender portal](#) → [Incidents & alerts](#) → [Alerts](#), filtered to the VM. Alerts should appear within roughly 5–15 minutes.

4. Phase 2 — Credential Access Simulation

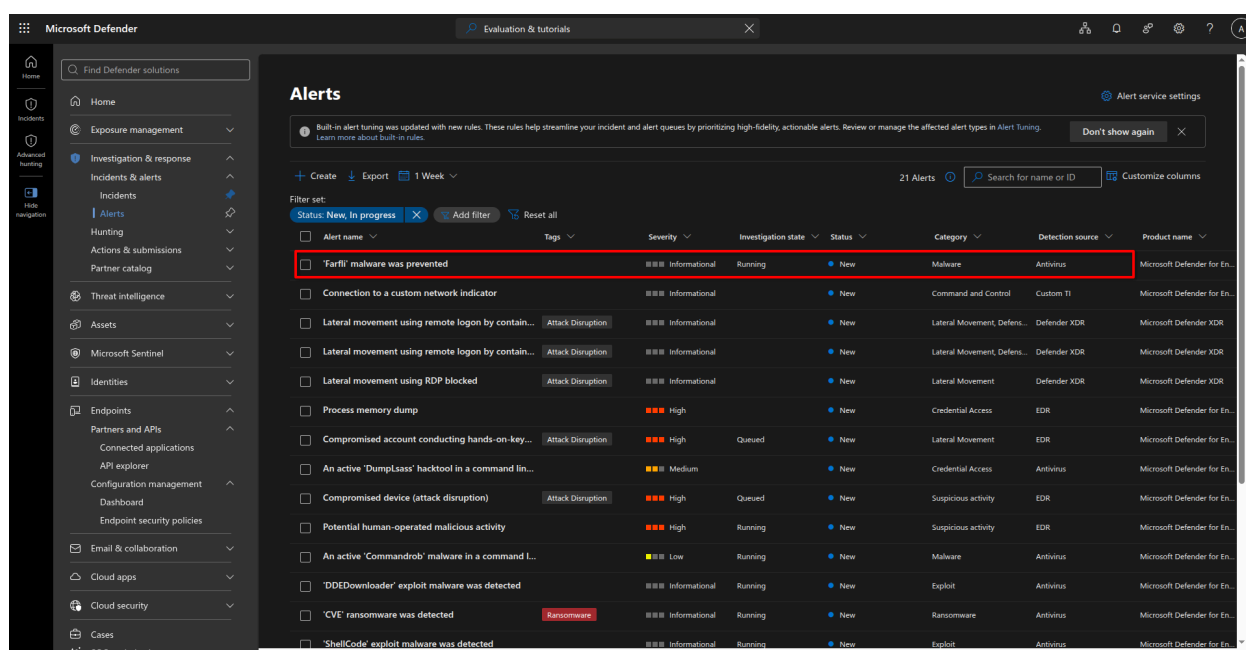
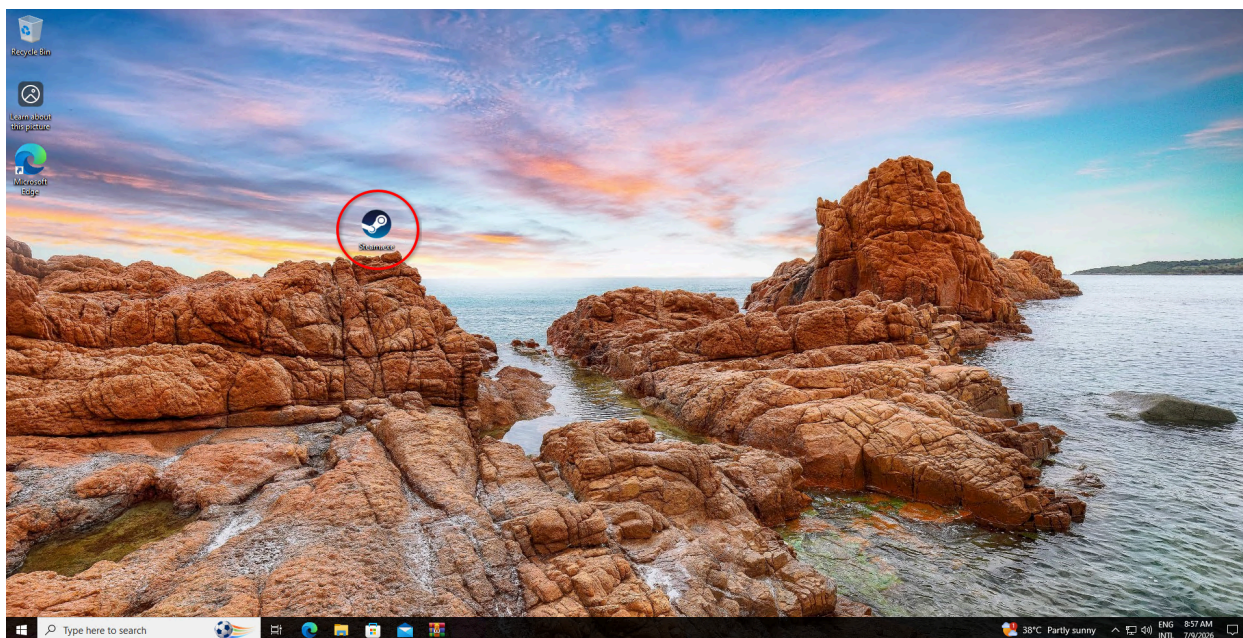
Pick RDP or SSH depending on which service is exposed.

```
# RDP brute force
hydra -L userlist.txt -P passwordlist.txt rdp://<azure_vm_ip>
```

Verify: In the same Alerts view, confirm a credential-access alert appears and begins linking to the same incident as Phase 1. Defender automatically groups related alerts into a single incident — check the incident's **Attack story** tab to confirm correlation.

Note: if a lateral-movement or execution-stage phase sits between credential access and initial access in your run, insert it here as Phase 3 before renumbering the section below.

1. Log in to the Azure VM over RDP using the credentials validated in Phase 2.
2. Deliver and execute the payload — in this run, a RAT disguised as a Steam-related executable.
3. Defender AV blocked the file on first execution attempt; this was logged as an alert and automatically kicked off an investigation.

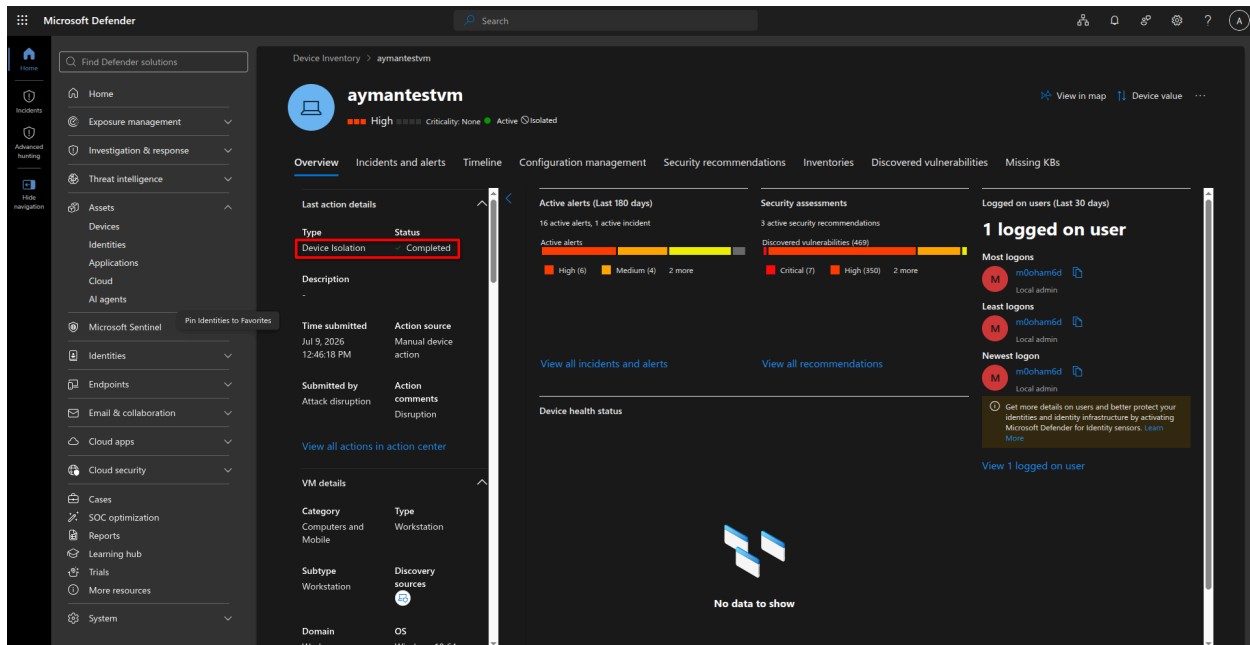
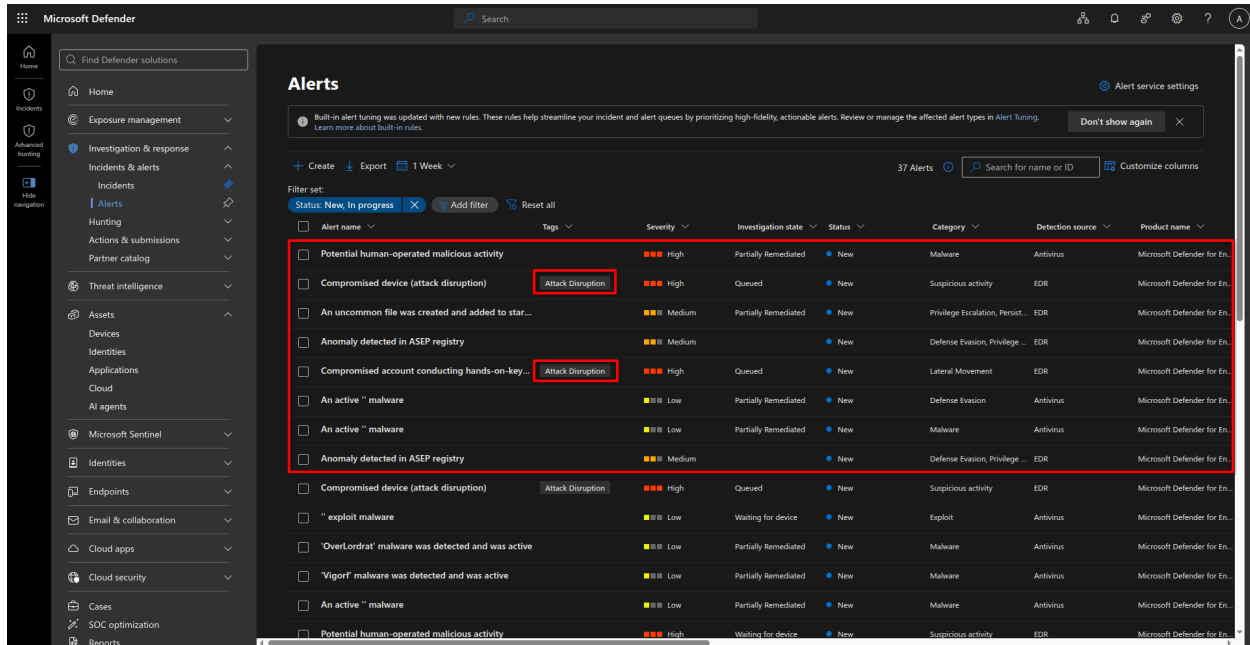


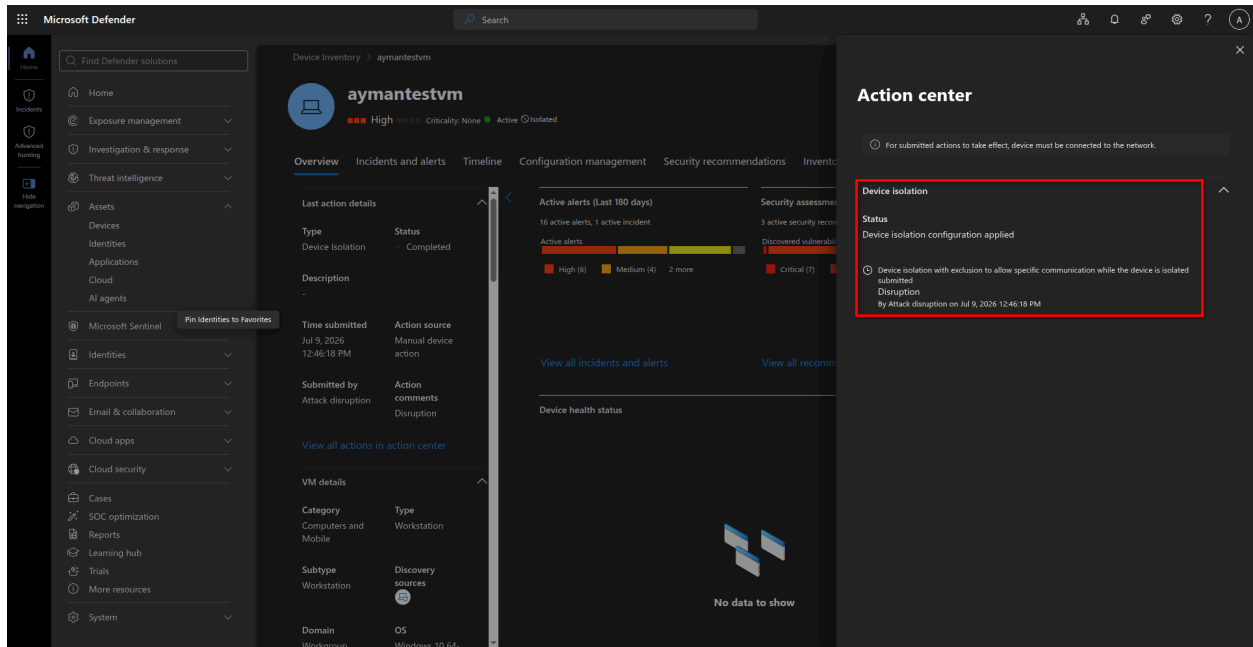
Verify: Defender portal → Incidents & alerts — confirm a new alert tied to the blocked execution appears and merges into the ongoing incident.

6. Result — Attack Disruption Triggered

With recon, credential access, and blocked execution all correlated into one incident, Defender applied attack disruption:

- The incident was tagged **Attack Disruption**.
- The Azure VM was automatically **isolated** from the network.



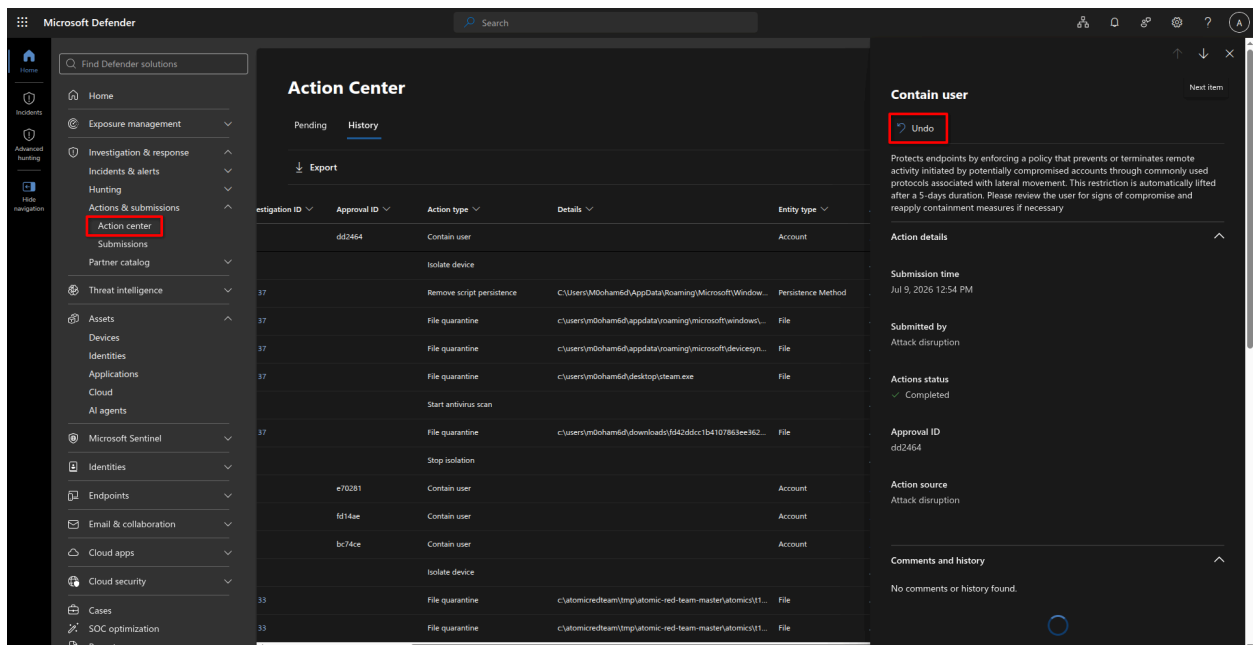


This confirms device containment fired as expected for the given signal set (see Section 1 — Scope).

7. Rolling Back Attack Disruption Actions

To undo an automated containment action:

Defender portal → Action Center → History → select the action → Undo



Summary

Phase	Action	Signal Generated
0	Prerequisite checks	N/A — environment validation
1	<code>nmap</code> scan from Kali	Reconnaissance / discovery alert
2	<code>hydra</code> brute force (RDP/SSH)	Credential access alert
3	RDP login + payload execution	Blocked-execution alert, investigation triggered
Result	Incident correlation	Attack Disruption tag + device isolation
Rollback	Action Center → History	Manual undo of containment